

Group 45

Lab 11 - State Machines

In this lab, you've learned about One Hot and Binary state machines and how to build them.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

We implemented a finite state machine using One Hot and Binary encoding methods. We learned how state transitions are managed using D flip-flops and to design and find the next state with logic. Through this lab we learned more about how coding impacts hardware complexity, debugging, and resource usage on an FPGA.

Lab Questions

1 - Compare and contrast One Hot and Binary encodings

One Hot:

- Each state has its own flip-flop
- Only one flip-flop is set at a time
- Easier to read, write, and debug
- Uses more flip-flops, more hardware resources

Binary

- States are represented as binary numbers using flip-flops
- More efficient way of using flip-flops, don't need one for each state
- Logic for the states is more complex
- Slightly hard to debug, but better when you have a lot of states

2 - Which method did your team find easier, and why?

We found One Hot encoding was easier to implement. The logic came more naturally and was more straightforward compared to binary. It was also visually simpler to trace which state we were in since each state has its own flip-flop.

3 - In what conditions would you have to use one over the other? Think about resource utilization on the FPGA.

You should use One Hot when you have a small number of states, want a simpler and easier logic, and you're optimizing for speed. You should use Binary encoding when you have a lot of states and slip-slop count matters. You should also use it if you want to optimize resource usage and use fewer flip-flops and when your design needs to fit into limited FPGA hardware.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.